

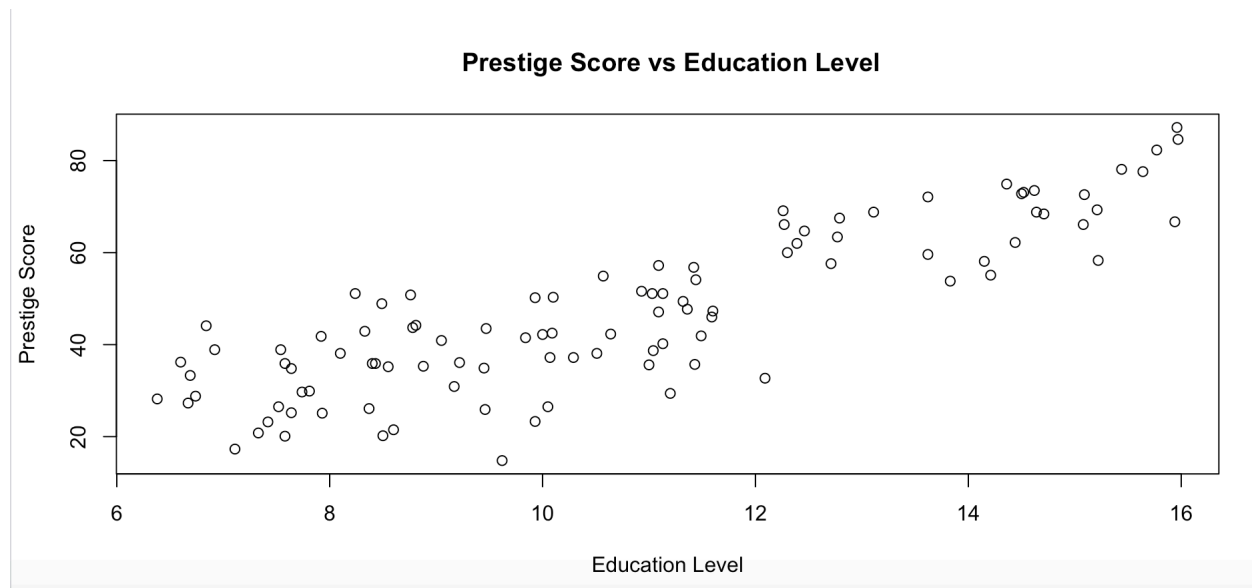
# MET CS 555 Assignment 4 – 20 points

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**SUBMISSION REQUIREMENTS:** Please submit a single document (word or PDF) for submission. Your submission should contain a summary of your results (and answers to questions asked on the homework) as well as your R code used to generate your results (please append to the end of your submission). Please use R for the calculations whenever possible. You will lose points if you are not utilizing R. You will also lose 10 points per day for late submissions unless prior arrangements are made with your facilitator.

**The data on the next two pages is from a Canadian 1970 census which collected information about specific occupations. Data collected was used to develop a regression model to predict prestige for all occupations. Use R to calculate the quantities and generate the visual summaries requested below.**

**(1)** To get a sense of the data, generate a scatterplot to examine the association between prestige score and years of education. Briefly describe the form, direction, and strength of the association between the variables. Calculate the correlation coefficient. **(3 points )**



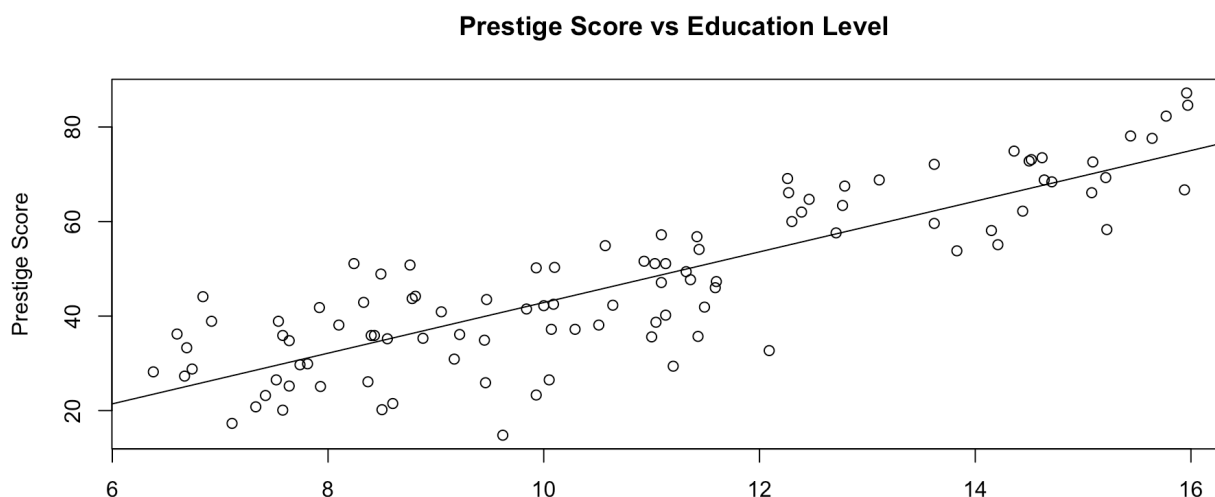
**Form:** the relationship appears to be linear.

**Direction:** The docs are scattered around the anti-diagonal, with more variance at the lower education levels.

**Strength of Association:** the more educated, the higher the prestige score.

correlation coefficient :0.85

(2) Perform a simple linear regression with prestige score and years of education, and briefly summarize your conclusions (no need to do the 5-step procedure here). Generate a residual plot. Assess whether the model assumptions are met. Are there any outliers or influence points? If so, identify them by ID and comment on the effect of each on the regression. (5 points)

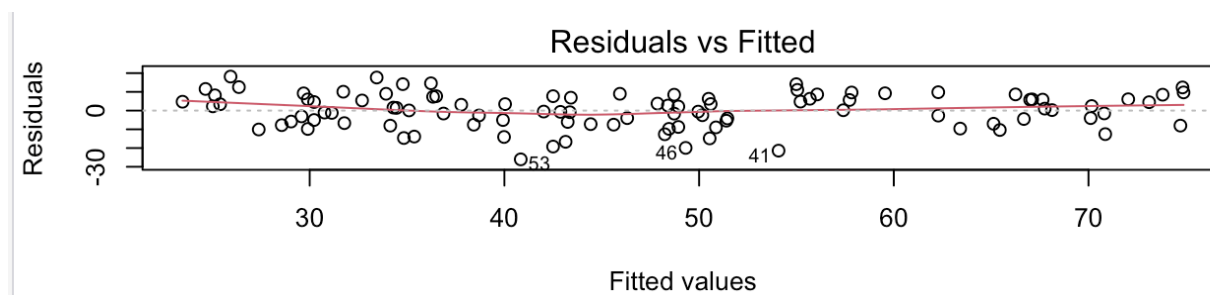


Coefficients:

|                            | Estimate | Std. Error | t value | Pr(> t ) |     |
|----------------------------|----------|------------|---------|----------|-----|
| (Intercept)                | -10.732  | 3.677      | -2.919  | 0.00434  | **  |
| dataframe\$Education.Level | 5.361    | 0.332      | 16.148  | < 2e-16  | *** |

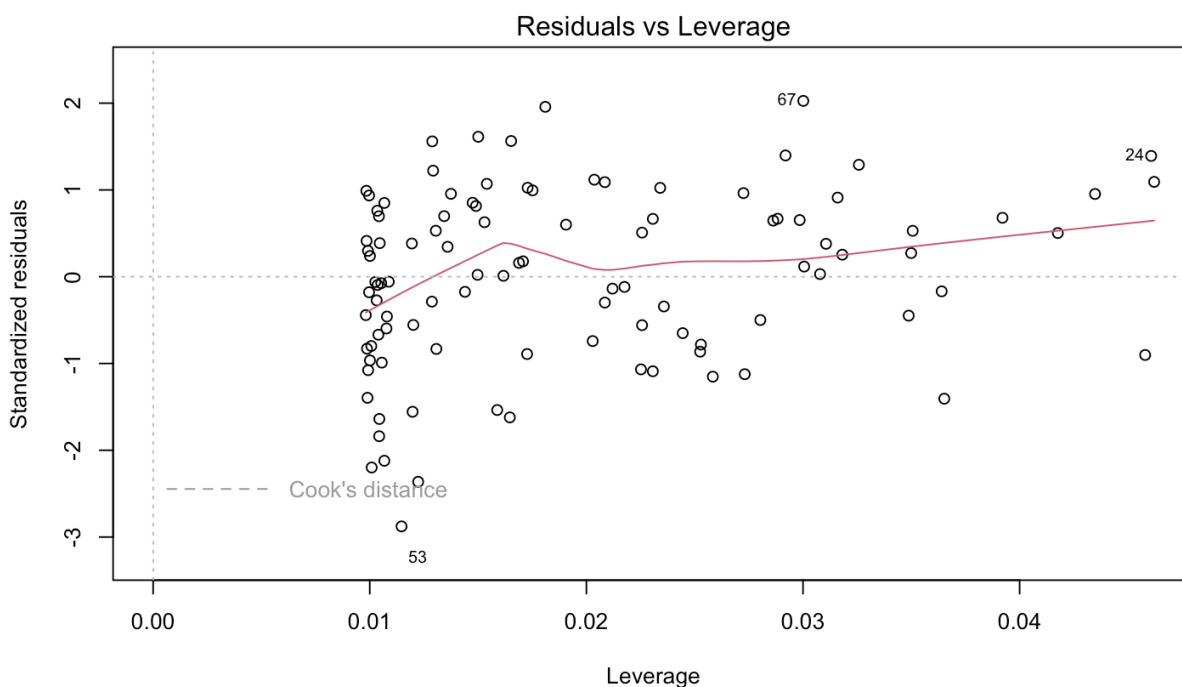
slope estimate: 5.361

the p-value for this estimate is very small, so we can reject the hypothesis that the slope is 0.



**Linearity:** there is a U shape in the LOESS line, this assumption is violated.

**Homoskedasticity:** it seems there is more variance of residuals in the range 30-50, so this assumption is violated.



**Influential Points:** 24,67,53 from above plot

**Effect of 24:** moderate residual, large leverage, pulls the line upward, increase the slope.

**Effect of 67:** large residual, moderate leverage, pulls the line upward, increasing the slope

**Effect of 53:** large negative residual, moderate leverage, pulls the line downward, increasing the slope.

**(3)** Calculate the least squares regression equation that predicts prestige score from education, income, and percentage of women. **Formally test** (using the 5-step procedure) whether the set of these predictors are associated with prestige score at the  $\alpha = 0.05$  level (Hint: You should be performing the global test). **(5 points)**

**Step1**

**H0:**  $\beta_1 = \beta_2 = \dots = \beta_n = 0$

H1: at least one of them is not equal to 0.

$\alpha = 0.05$

Step2

We choose f test since this is a global test.

Step3

if  $f_{\text{statistic}} \geq f_{\text{critical}}$ , we reject the null hypothesis.

Step4

$f_{\text{statistic}} = 129.2$

$f_{\text{critical}} = 2.697423$

Step 5

since  $f_{\text{statistic}} > f_{\text{critical}}$ , we reject the null hypothesis. In other words, we have significant data at level 0.05 showing that at least one of the betas is not 0.

(4) If the overall model was significant, summarize the information about the contribution of each variable separately at the same significance level as used for the overall model (*no need to do a formal 5-step procedure for each one, just comment on the results of the tests*). Provide interpretations for any estimates (of the slopes) that are significant. Calculate 95% confidence intervals for any estimates that are significant. (4 points)

Coefficients:

|                                                | Estimate   | Std. Error | t value | Pr(> t )     |
|------------------------------------------------|------------|------------|---------|--------------|
| (Intercept)                                    | -6.7943342 | 3.2390886  | -2.098  | 0.0385 *     |
| dataframe\$Education.Level                     | 4.1866373  | 0.3887013  | 10.771  | < 2e-16 ***  |
| dataframe\$Income                              | 0.0013136  | 0.0002778  | 4.729   | 7.58e-06 *** |
| dataframe\$Percent.of.Workforce.that.are.Women | -0.0089052 | 0.0304071  | -0.293  | 0.7702       |

Education level: the contribution is significant as the p value is less than 0.05.

Income: the contribution is significant as the p value is less than 0.05.

Percent of woman: p value is larger than 0.05 so we fail to reject the null hypothesis that the beta for this predictor is 0. As a result, the contribution from this predictor is little.

Interpretations:

Education level: for every one unit increase in education level, the prestige score increases by 4.19

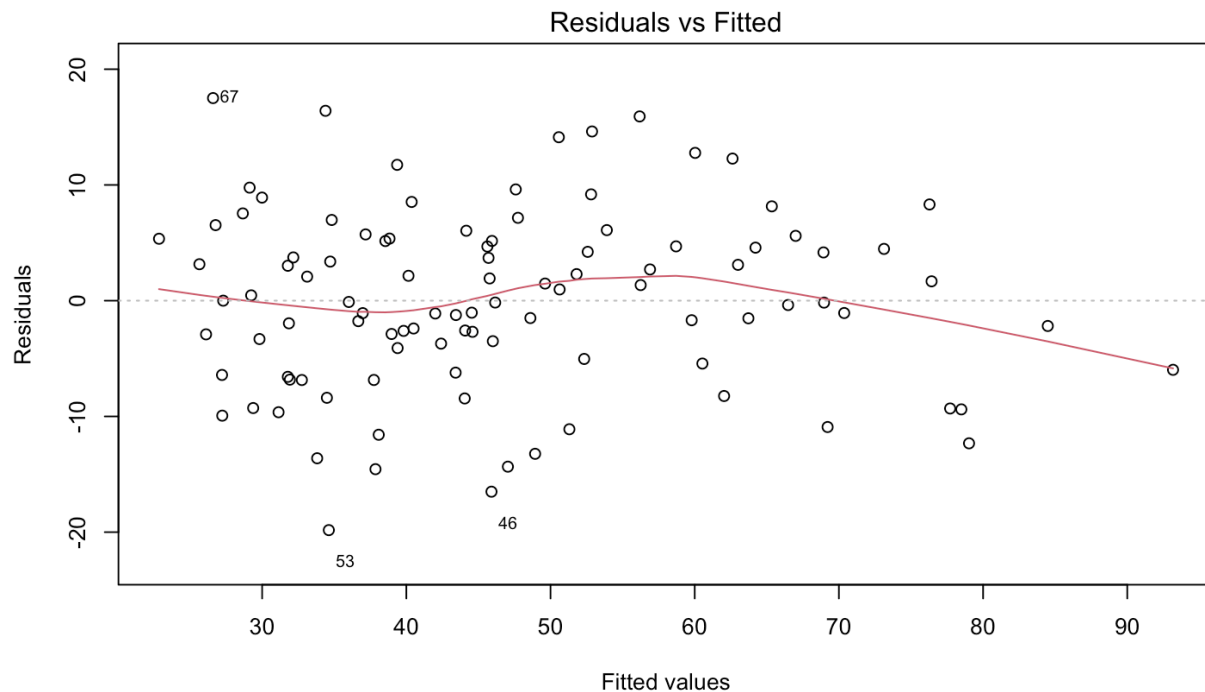
Income: for every one unit increase in income, the prestige score increases by 0.001.

**Confidence Interval:**

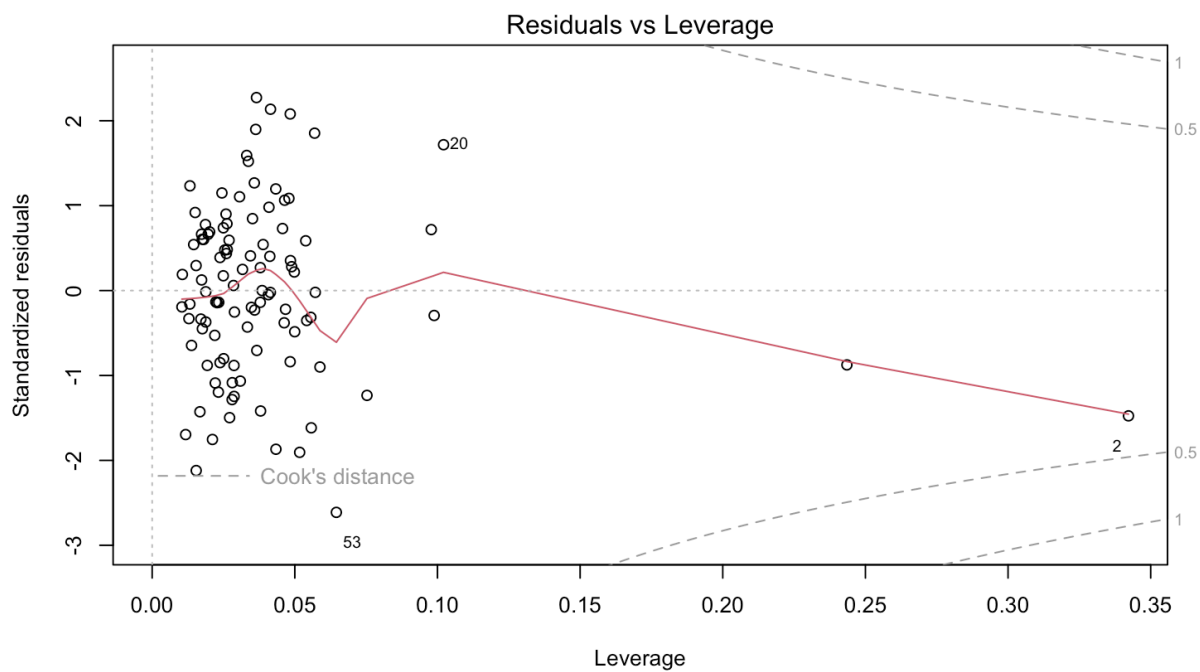
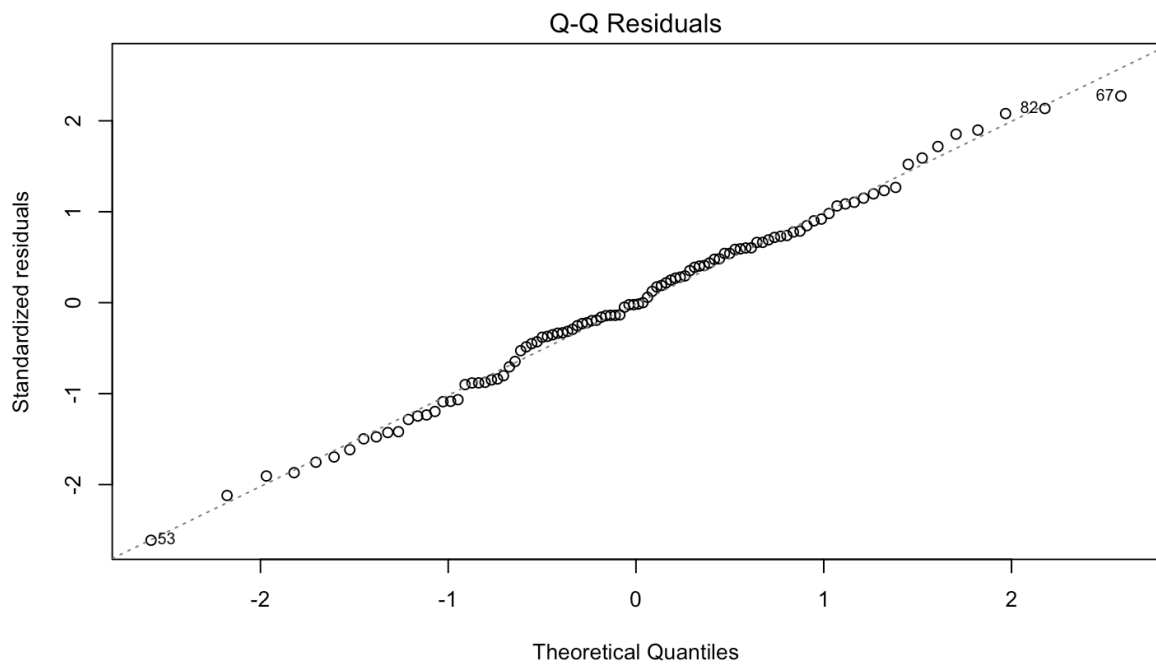
|                                                | 2.5 %         | 97.5 %       |
|------------------------------------------------|---------------|--------------|
| (Intercept)                                    | -1.322220e+01 | -0.366468202 |
| dataframe\$Education.Level                     | 3.415272e+00  | 4.958002277  |
| dataframe\$Income                              | 7.623127e-04  | 0.001864808  |
| dataframe\$Percent.of.Workforce.that.are.Women | -6.924697e-02 | 0.051436660  |

**Education Level [3.42,4.96]****Income: [0.0007,0.0018]**

(5) Generate a residual plot showing the fitted values from the regression against the residuals. Is the fit of the model reasonable? Are there any outliers or influence points? **(3 points)**



The plot above shows that the fit is NOT reasonable as the LOESS line has a wave- like structure. In addition, it seems the variance of residuals decreases as the fitted value increases.



influential points: 2,20,53

outliers: none because no point lies beyond the Cook's distance contour = 1.

# Data Set for Assignment 4

|  Occupational Title | Education Level (years) | Income (\$) | Percent of Workforce that are Women | Prestige Score |
|------------------------------------------------------------------------------------------------------|-------------------------|-------------|-------------------------------------|----------------|
| GOV. ADMINISTRATORS                                                                                  | 13.11                   | 12351       | 11.16                               | 68.8           |
| GENERAL MANAGERS                                                                                     | 12.26                   | 25879       | 4.02                                | 69.1           |
| ACCOUNTANTS                                                                                          | 12.77                   | 9271        | 15.7                                | 63.4           |
| PURCHASING OFFICERS                                                                                  | 11.42                   | 8865        | 9.11                                | 56.8           |
| CHEMISTS                                                                                             | 14.62                   | 8403        | 11.68                               | 73.5           |
| PHYSICISTS                                                                                           | 15.64                   | 11030       | 5.13                                | 77.6           |
| BIOLOGISTS                                                                                           | 15.09                   | 8258        | 25.65                               | 72.6           |
| ARCHITECTS                                                                                           | 15.44                   | 14163       | 2.69                                | 78.1           |
| CIVIL ENGINEERS                                                                                      | 14.52                   | 11377       | 1.03                                | 73.1           |
| MINING ENGINEERS                                                                                     | 14.64                   | 11023       | 0.94                                | 68.8           |
| SURVEYORS                                                                                            | 12.39                   | 5902        | 1.91                                | 62             |
| DRAUGHTSMEN                                                                                          | 12.3                    | 7059        | 7.83                                | 60             |
| COMPUTER PROGRAMERS                                                                                  | 13.83                   | 8425        | 15.33                               | 53.8           |
| ECONOMISTS                                                                                           | 14.44                   | 8049        | 57.31                               | 62.2           |
| PSYCHOLOGISTS                                                                                        | 14.36                   | 7405        | 48.28                               | 74.9           |
| SOCIAL WORKERS                                                                                       | 14.21                   | 6336        | 54.77                               | 55.1           |
| LAWYERS                                                                                              | 15.77                   | 19263       | 5.13                                | 82.3           |
| LIBRARIANS                                                                                           | 14.15                   | 6112        | 77.1                                | 58.1           |
| VOCATIONAL COUNSELLORS                                                                               | 15.22                   | 9593        | 34.89                               | 58.3           |
| MINISTERS                                                                                            | 14.5                    | 4686        | 4.14                                | 72.8           |
| UNIVERSITY TEACHERS                                                                                  | 15.97                   | 12480       | 19.59                               | 84.6           |
| PRIMARY SCHOOL TEACHERS                                                                              | 13.62                   | 5648        | 83.78                               | 59.6           |
| SECONDARY SCHOOL TEACHERS                                                                            | 15.08                   | 8034        | 46.8                                | 66.1           |
| PHYSICIANS                                                                                           | 15.96                   | 25308       | 10.56                               | 87.2           |
| VETERINARIANS                                                                                        | 15.94                   | 14558       | 4.32                                | 66.7           |
| OSTEOPATHS CHIROPRACTORS                                                                             | 14.71                   | 17498       | 6.91                                | 68.4           |
| NURSES                                                                                               | 12.46                   | 4614        | 96.12                               | 64.7           |
| NURSING AIDES                                                                                        | 9.45                    | 3485        | 76.14                               | 34.9           |
| PHYSIO THERAPSTS                                                                                     | 13.62                   | 5092        | 82.66                               | 72.1           |
| PHARMACISTS                                                                                          | 15.21                   | 10432       | 24.71                               | 69.3           |
| MEDICAL TECHNICIANS                                                                                  | 12.79                   | 5180        | 76.04                               | 67.5           |
| COMMERCIAL ARTISTS                                                                                   | 11.09                   | 6197        | 21.03                               | 57.2           |
| RADIO TV ANNOUNCERS                                                                                  | 12.71                   | 7562        | 11.15                               | 57.6           |
| ATHLETES                                                                                             | 11.44                   | 8206        | 8.13                                | 54.1           |
| SECRETARIES                                                                                          | 11.59                   | 4036        | 97.51                               | 46             |
| TYPISTS                                                                                              | 11.49                   | 3148        | 95.97                               | 41.9           |
| BOOKKEEPERS                                                                                          | 11.32                   | 4348        | 68.24                               | 49.4           |
| TELLERS CASHIERS                                                                                     | 10.64                   | 2448        | 91.76                               | 42.3           |
| COMPUTER OPERATORS                                                                                   | 11.36                   | 4330        | 75.92                               | 47.7           |
| SHIPPING CLERKS                                                                                      | 9.17                    | 4761        | 11.37                               | 30.9           |
| FILE CLERKS                                                                                          | 12.09                   | 3016        | 83.19                               | 32.7           |
| RECEPTIONSTS                                                                                         | 11.04                   | 2901        | 92.86                               | 38.7           |
| MAIL CARRIERS                                                                                        | 9.22                    | 5511        | 7.62                                | 36.1           |
| POSTAL CLERKS                                                                                        | 10.07                   | 3739        | 52.27                               | 37.2           |
| TELEPHONE OPERATORS                                                                                  | 10.51                   | 3161        | 96.14                               | 38.1           |
| COLLECTORS                                                                                           | 11.2                    | 4741        | 47.06                               | 29.4           |
| CLAIM ADJUSTORS                                                                                      | 11.13                   | 5052        | 56.1                                | 51.1           |
| TRAVEL CLERKS                                                                                        | 11.43                   | 6259        | 39.17                               | 35.7           |
| OFFICE CLERKS                                                                                        | 11                      | 4075        | 63.23                               | 35.6           |
| SALES SUPERVISORS                                                                                    | 9.84                    | 7482        | 17.04                               | 41.5           |
| COMMERCIAL TRAVELLERS                                                                                | 11.13                   | 8780        | 3.16                                | 40.2           |
| SALES CLERKS                                                                                         | 10.05                   | 2594        | 67.82                               | 26.5           |
| NEWSBOYS                                                                                             | 9.62                    | 918         | 7                                   | 14.8           |
| SERVICE STATION ATTENDANT                                                                            | 9.93                    | 2370        | 3.69                                | 23.3           |
| INSURANCE AGENTS                                                                                     | 11.6                    | 8131        | 13.09                               | 47.3           |
| REAL ESTATE SALESMEN                                                                                 | 11.09                   | 6992        | 24.44                               | 47.1           |
| BUYERS                                                                                               | 11.03                   | 7956        | 23.88                               | 51.1           |
| FIREFIGHTERS                                                                                         | 9.47                    | 8895        | 0                                   | 43.5           |
| POLICEMEN                                                                                            | 10.93                   | 8891        | 1.65                                | 51.6           |

|                        |       |       |       |      |
|------------------------|-------|-------|-------|------|
| COOKS                  | 7.74  | 3116  | 52    | 29.7 |
| BARTENDERS             | 8.5   | 3930  | 15.51 | 20.2 |
| FUNERAL DIRECTORS      | 10.57 | 7869  | 6.01  | 54.9 |
| BABYSITTERS            | 9.46  | 611   | 96.53 | 25.9 |
| LAUNDERERS             | 7.33  | 3000  | 69.31 | 20.8 |
| JANITORS               | 7.11  | 3472  | 33.57 | 17.3 |
| ELEVATOR OPERATORS     | 7.58  | 3582  | 30.08 | 20.1 |
| FARMERS                | 6.84  | 3643  | 3.6   | 44.1 |
| FARM WORKERS           | 8.6   | 1656  | 27.75 | 21.5 |
| ROTARY WELL DRILLERS   | 8.88  | 6860  | 0     | 35.3 |
| BAKERS                 | 7.54  | 4199  | 33.3  | 38.9 |
| SLAUGHTERERS 1         | 7.64  | 5134  | 17.26 | 25.2 |
| SLAUGHTERERS 2         | 7.64  | 5134  | 17.26 | 34.8 |
| CANNERS                | 7.42  | 1890  | 72.24 | 23.2 |
| TEXTILE WEAVERS        | 6.69  | 4443  | 31.36 | 33.3 |
| TEXTILE LABOURERS      | 6.74  | 3485  | 39.48 | 28.8 |
| TOOL DIE MAKERS        | 10.09 | 8043  | 1.5   | 42.5 |
| MACHINISTS             | 8.81  | 6686  | 4.28  | 44.2 |
| SHEET METAL WORKERS    | 8.4   | 6565  | 2.3   | 35.9 |
| WELDERS                | 7.92  | 6477  | 5.17  | 41.8 |
| AUTO WORKERS           | 8.43  | 5811  | 13.62 | 35.9 |
| AIRCRAFT WORKERS       | 8.78  | 6573  | 5.78  | 43.7 |
| ELECTRONIC WORKERS     | 8.76  | 3942  | 74.54 | 50.8 |
| RADIO TV REPAIRMEN     | 10.29 | 5449  | 2.92  | 37.2 |
| SEWING MACH OPERATORS  | 6.38  | 2847  | 90.67 | 28.2 |
| AUTO REPAIRMEN         | 8.1   | 5795  | 0.81  | 38.1 |
| AIRCRAFT REPAIRMEN     | 10.1  | 7716  | 0.78  | 50.3 |
| RAILWAY SECTIONMEN     | 6.67  | 4696  | 0     | 27.3 |
| ELECTRICAL LINEMEN     | 9.05  | 8316  | 1.34  | 40.9 |
| ELECTRICIANS           | 9.93  | 7147  | 0.99  | 50.2 |
| CONSTRUCTION FOREMEN   | 8.24  | 8880  | 0.65  | 51.1 |
| CARPENTERS             | 6.92  | 5299  | 0.56  | 38.9 |
| MASONS                 | 6.6   | 5959  | 0.52  | 36.2 |
| HOUSE PAINTERS         | 7.81  | 4549  | 2.46  | 29.9 |
| PLUMBERS               | 8.33  | 6928  | 0.61  | 42.9 |
| CONSTRUCTION LABOURERS | 7.52  | 3910  | 1.09  | 26.5 |
| PILOTS                 | 12.27 | 14032 | 0.58  | 66.1 |
| TRAIN ENGINEERS        | 8.49  | 8845  | 0     | 48.9 |
| BUS DRIVERS            | 7.58  | 5562  | 9.47  | 35.9 |
| TAXI DRIVERS           | 7.93  | 4224  | 3.59  | 25.1 |
| LONGSHOREMEN           | 8.37  | 4753  | 0     | 26.1 |
| TYPESETTERS            | 10    | 6462  | 13.58 | 42.2 |
| BOOKBINDERS            | 8.55  | 3617  | 70.87 | 35.2 |

```
dataframe = read.csv("assignment4data.csv")
```

```
par(mfrow = c(1,1))
```

```
#question1
```

```
plot(
```

```
  dataframe$Education.Level,
```

```
  dataframe$Prestige.Score,
```

```
  ylab = "Prestige Score",
```

```
  xlab = "Education Level",
```

```
  main = "Education Level vs Prestige Score"
```



```

)

cor(dataframe$Prestige.Score,dataframe$Education.Level)

# question 2

model1 = lm(dataframe$Prestige.Score ~ dataframe$Education.Level)

abline(model1,col ="red",lwd = 2)

plot(
  model1$fitted.values,
  model1$residuals,
  xlab = "Prestige Score",
  ylab = "Residual",
  main = "Residual Vs Prestige Score"
)

abline(h=0)

par(mfrow =c(2,2))

plot(model1)

# question3 and 4

model3 = lm(dataframe$Prestige.Score ~
  dataframe$Education.Level +
  dataframe$Income +

```

```
dataframe$Percent.of.Workforce.that.are.Women)  
  
summary(model3)  
  
alpha = 0.05  
  
n = nrow(dataframe)  
  
k = 3  
  
f_critical = qf(1-alpha,k,n-k-1)  
  
f_critical  
  
confint(model3)  
  
# question 5  
  
plot(model3)
```